

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

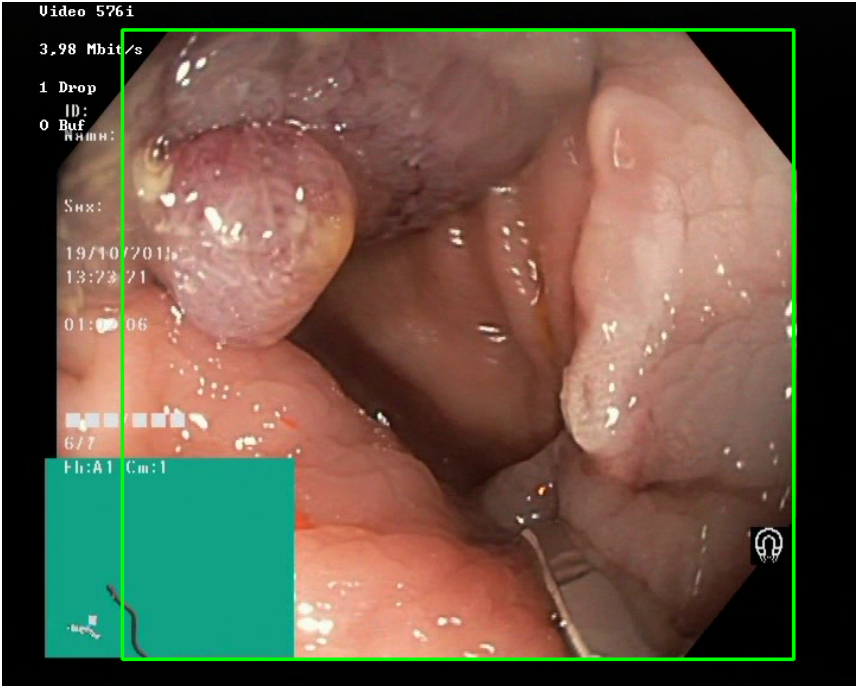
Video ID:	7e3731a7-51f6-4415-9038-41322f6e4746
Total Frames:	10180
Frame Rate:	25.0 fps (assumed)
Duration:	407.2 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	4467	✓
RT-DETR Detections	6074	✓
MedSAM2 Detections	6445	✓
Consensus (3-Model Agreement)	2	✓
Risk Classification	Count	Percentage
HIGH RISK	1	50.0%
MEDIUM RISK	1	50.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	2	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — SERRATED_POLYP [MEDIUM_RISK] (Tracked across 692 frames)



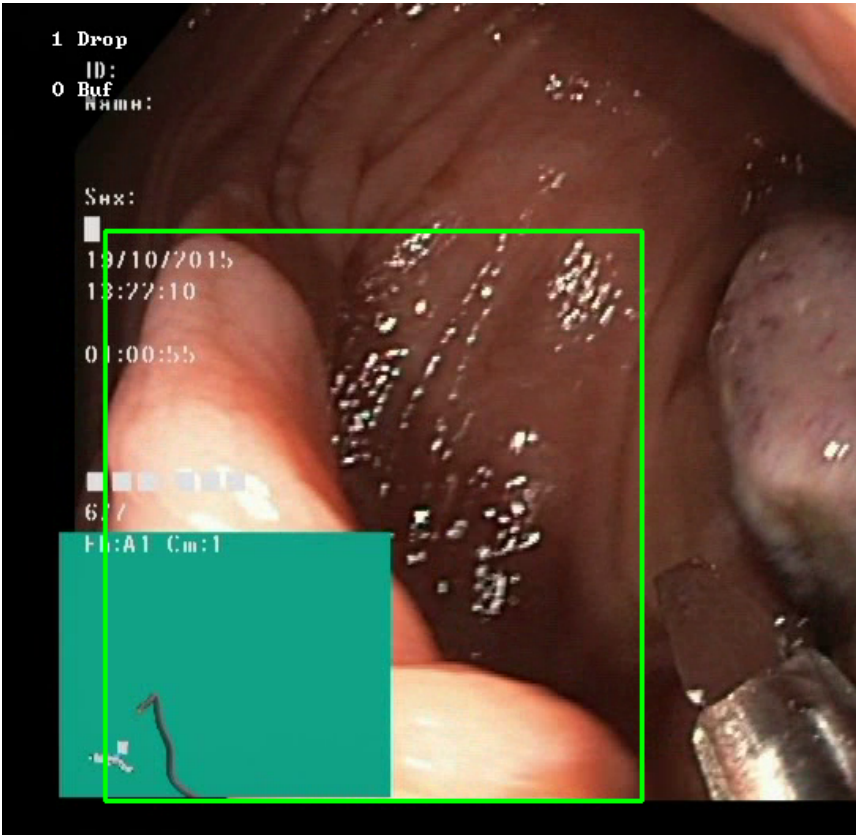
Feature	Value	Interpretation
Frame Index	6804	Frame 6804 of 10180
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	SERRATED_POLYP	Type confidence: 92.0%
Redness Score	0.139	Color-based vascularization
Relative Radius	47.6%	Polyp size as % of frame diagonal
Texture Score	0.702	Surface morphology
Vessel Visibility	0.310	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	85.9%	YOLO / RT-DETR / track confidence

Type Assessment: Serrated lesion — consider serrated polyposis syndrome. Complete resection required.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #2 — MALIGNANT_POLYP [HIGH_RISK] (Tracked across 2308 frames)



Feature	Value	Interpretation
Frame Index	5013	Frame 5013 of 10180
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	MALIGNANT_P OLYP	Type confidence: 92.0%
Redness Score	0.141	Color-based vascularization
Relative Radius	54.9%	Polyp size as % of frame diagonal

Texture Score	0.171	Surface morphology
Vessel Visibility	0.488	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	88.9%	YOLO / RT-DETR / track confidence

Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.

Low-confidence HIGH RISK detection - Further evaluation recommended

CLINICAL IMPRESSION

Summary:

A total of 2 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 1 (50.0%)
- MEDIUM RISK polyps: 1 (50.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

The presence of HIGH RISK polyps warrants close clinical attention. Consider endoscopic resection or follow-up surveillance depending on clinical context.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.